

## RESEARCH ARTICLE

# CrediBot: Applying Bot Detection for Credibility Analysis on Twitter

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**ABSTRACT** Nowadays, people and organizations use social networks for allowing and facilitating the transfer of information among groups that share similar interests. Due to the wide repertoire of users that these social platforms have and the amount of information generated within them, the presence of bots has become a relevant issue, both to facilitate the sharing of true information or to disseminate false information (fake news). In the second case, bots could manipulate political opinions, be perpetrators of identity or information theft, among other possible dangers that can cause when interacting on the platform. Thus, the identification of bots in social networks can become a useful practice to evaluate credibility or to detect fake news. In this work, we extend a previously proposed credibility model for Twitter, by incorporating bot detection. The original model calculates the credibility of tweets based on three measures: text, account/user, and social impact, using different filters to analyse text (SPAM, bad words, and good spelling) and account attributes (e.g., creation date, followers, following) to calculate account/user and social credibility scores. The extended model considers in the user credibility, the bot verification. Additionally, the extended credibility model is implemented in T-CREo, a framework for real time credibility analysis. For bot detection, some machine learning algorithms for supervised learning, such as AdaBoost, Bagging, Decision Tree, Logistic Regression, and Random Forest are trained and evaluated. Results show that the best algorithm is the Random Forest for its capacity of generalization with an accuracy and F1-score values over 97% both in English and Spanish. The evaluation of the bot detection functionality in the credibility analysis shows a performance of precision=1.0, recall=0.8462, F1-score=0.9167, and accuracy=0.92 for both English and Spanish models in our validation tests.

**INDEX TERMS** Credibility analysis, bot detection, social networks, twitter.

## I. INTRODUCTION

Currently, social networks are the main means of communication around the world, daily generating a huge amount of information through the interaction of the people who use these platforms. In these interactions, people share all kinds of information about themselves or other users, and express their opinions and points of view on any topic of public or private interest [1]. Social networks have different purposes depending on their main use. For instance,

Facebook generates relationships among users; YouTube is for entertainment activities; and LinkedIn is used for more professional contexts [2]. Some of the most recognized social networks are Facebook and Twitter, being the latter the most used for data analysis due to the simplicity to obtain the data (through an API) and its format [3].

Because these social networks are the main source of information for many users, they have become a main focus of interest for the presence of “social bots”. A “bot” is defined as an automated program for interacting on a social network, with the goal of mimicking the behavior of a human user [4]. While useful bots exist within social networks (reminders,

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archivers, weather reports, etc.), several of these bots are used for harmful actions within the platform. Shao et al. [5] show evidence that social bots are an effective tool to manipulate social media and they are useful to spread low-credibility content, such as false news, conspiracy theories, and junk science. Malicious bots can be used to artificially increase the popularity of a person or social movement or group [6]; to influence or manipulate people for political purposes [7]; to manipulate financial markets [8], [9]; to disseminate spam or false information [10], [11], [12]; and for conducting fraud or theft of other users' information [13], [14], [15].

Works on bot detection make different classifications regarding bots, differentiating them by their behavior, usefulness, or way of operating [4], [16]. Some of these classifications include:

- **Consumer Bots:** Bots whose main objective is to obtain content from various sources and provide update services, such as weather reports, horoscope, or news for consumption or personal use [4].
- **Broadcast Bots:** Bots with the main purpose of disseminating information to a general audience, through news articles, sites of interest, or events. They are usually used and managed by organizations or groups of people. Their content is usually benign [4].
- **Spam bots:** These bots are responsible for disseminating content with false information, or harmless, or irrelevant content on a repetitive basis. They also redirect to malicious sites for fraud. These bots are considered malicious [4].
- **Hybrid Bots or Cyborgs:** This classification corresponds to user accounts that are managed by human users with the help or support of a bot or vice versa [16].
- **Crawlers and scrapers:** They are bots that have been designed to index and archive webpages, for example, to make the websites available through search engines.
- **Chatbots:** They are computer programs that converse with people in real-time using natural language. Frequently used by businesses for easier customer interaction, such as responding to frequently asked queries.
- **Spambots:** They are software or computer systems that have been compromised by malware and are under the control of a third party. They are used to send out a large number of messages, advertising, comments, etc.
- **Social bots:** Active on social media, these programs create content on their own and frequently pass for genuine people.
- **Sockpuppets:** They are genuine people who assume false identities, often to advance a certain cause or product, such as business promotion, favorable book, and film reviews. The intention of these sockpuppets is to create opinion bias towards an entity and usually they are employed for online deception [17].

Considering this classification, it is important to highlight that not all bots found in social networks, in this case Twitter, correspond to programs designed with bad intentions. Many

useful bots are used by several real users to improve the quality of experience within this platform or with beneficial purposes [11]. It is also important to take into account the existence of hybrid bots or cyborgs, since the pattern of activity that these bots have are totally different from those of a common bot or a human user, making their correct identification even more difficult [11].

Twitter is a very popular microblogging service with structural characteristics of both an information network and a social network [18]. An evidence of this popularity is the fact that the number of Twitter accounts have doubled in the last 5 years passing of 109 millions in 2017 to 237 millions in 2022.<sup>1</sup> Nevertheless, the users are exposed to fake news or being manipulated with any type of interest. A study in 2017 declares that around 15% of Twitter accounts would correspond to bots [19], possibly mirroring the situation of most other social networks on the web [20]. Many works have been interested in the issue of bot detection and there is a clear need for the research community to develop new technologies that can identify social bots [21], however this remains still a challenge [20], [22], [23]. On the other hand, there is also a latent requirement for automatic systems to help users verify the reliability/credibility of the sources they access. In the current era of information overload, restlessness, and uncertainty, it is essential to create models for confirming information from online social networks like Twitter [24]. When using the richness of social information available about almost anything, trust plays a crucial part in assisting social network users in making the right choices [25].

In this context, this work extends a credibility model for Twitter presented in [26], by incorporating bot detection. The original model analyses the credibility of a tweet considering tree dimensions: text, account/user, and social impact. The new extended model considers, within the user credibility, the identification of bots, in order to detect the nature of the analyzed user. Moreover, in order to evaluate the accuracy of the extended credibility model, it is incorporated into T-CREo, a framework to perform credibility analysis in Twitter, in real time [27]. In summary, the main contributions of this work are summarized as follows:

- Two datasets for bot detection (English and Spanish), extending the ones proposed in [28] with content features, such as wavelet average and entropy;
- An algorithm for Twitter bots detection in real time, based on predictive models; one for tweets in English and the other one for tweets in Spanish;
- A comparative study for identifying the best features involved in the models that characterize the bots;
- An extended credibility model that integrates a bot detection function to calculate the level of credibility of tweets, incorporated into T-CREo to demonstrate its functionality and suitability.

<sup>1</sup><https://www.statista.com/statistics/970920/monetizable-daily-active-twitter-users-worldwide/>

The remainder of this paper is organized as follows. Section II corresponds to the review of works related to bot detection in Twitter and some works which consider characteristics of the user of a Twitter account, in their credibility analysis. Section III describes the methods used for Twitter bot detection including the dataset configuration, the preprocessing, and the training and selection of predictive models. In Section IV, the original credibility model proposed in [26] is described. Section V presents the addition of bot detection in the extended credibility model. Finally, Section VI discuss the main results and the future work.

## II. RELATED WORK

In this section, some studies are presented and analyzed mainly according to the bot detection implementation, regardless of the prediction objective it fulfills (i.e., detection of benign bots, malicious bots, or cyborgs). Additionally, some works focused on credibility in Twitter, that consider characteristics of users are summarized.

### A. DETECTION OF BOTS ON TWITTER

Bot detection in social networks is a topic that has been researched and still continues to be a challenge. Some reviews of the most recent techniques that have been published analyse the different classifiers used, the type of dataset used, and the features selected, as well as presenting comparisons among the evaluation techniques used for their validation [21], [29], [30], [31]. Alothali et al. [21] identify three common methods, based on graph, crowdsourcing, and machine learning, for social bot detection on social networks. For the machine learning algorithms, authors select 20 features of content, profile, and behavior from accounts on social networks. Orabi et al. [29] elaborate a bibliographic review of published works between 2010 and 2019 resulting in a taxonomy of detection methods and techniques used to detect bots in social media and a comparison among them. Aljabri et al. [30] survey existing studies from 2015 to 2022, which apply supervised, semi-supervised, and unsupervised machine learning techniques for social media bot detection. Authors consider the detection of social bots, spambots, and sybil bots on Facebook, Instagram, LinkedIn, Twitter, and Weibo, and provide a novel taxonomy of social media bot detection using machine learning based techniques. Moreover, they present an analysis of the most commonly extracted features and datasets used on each social media platform. Beatson et al. [31] compare a semi-automated method based on fixed criteria against a manual model based on open source intelligence (OSINT) to detect bots in Twitter. The evaluation is performed in terms of accuracy, user friendliness, efficiency or time costs, and simplicity of operation. Authors conclude that semi-automated method removes the need for manual checking, and a manual approach relies on searching for unusual patterns of behavior.

Some proposals consider machine learning approaches using supervised algorithms [16], [32], [33], [34], [35], [36],

[37]. Chu et al. [16] focus their work on the classification of humans, bots, and cyborgs on Twitter. They consider a user's content posting time period as a feature to be analyzed, where a lower entropy value corresponds to an apparent sign of automation, while a higher entropy value would be reflecting human involvement in the account activity. Authors conclude that, due to the more defined behavioral pattern of real users and bots, both are easier to identify than cyborgs. Indeed, the classification of cyborgs can yield many false positives considering their mixed behavior as humans and bots. The accuracy of their classification model reaches 96% for binary prediction (bots and humans). Alarifi et al. [32] propose a supervised classifier considering bots, hybrids, and human users as the three prediction categories. Authors performed data acquisition through Twitter API REST and Twitter Streaming API collecting information about 1.8 million accounts. They then performed labeling of the data by a selected group of people trained in classification. After the tests performed, they obtained 96% of accuracy and a Kappa coefficient at 0.61, resulting in a good percentage of agreement among the labeling group. They use four learning algorithms for training: Random Forest, BayesNet, Support Vector Machine (SVM), and multilayer artificial neural network. After the tests carried out, the predictive models obtained by Random Forest and BayesNet proved to have an accuracy percentage of 91%. Kudugunta et al. [33] describe a work using a neural network based on contextual short-term memory (LSTM), where both tweet content and metadata are considered. Social features of the user are also used, taking into account only the detection of bots at the tweet level, obtaining an accuracy of 96%.

Barbon et al. [34] propose an algorithm based on Discrete Wavelet Transform to obtain a pattern of writing style embedded in post contents in order to classify authors/users as being a human, a legitimate robot, or a malicious robot. Authors trained a classification model using the Random Forest algorithm over two real Twitter datasets (Super Bowl XLVIII event and Seattle Seahawks) and their model achieved an average accuracy of 94.47%. Hayawi et al. [35] propose a deep neural network designed with LSTM units and dense layers to classify human or bot accounts in Twitter. The classification is based on user profile metadata, including username, screen name, tweet count, followers count, friends count, listed count, user created date, description, location, url, verified flag, obtained through the Twitter API. The proposed model is evaluated on several datasets using two manners: testing on a hold-out set of the same dataset and training with one dataset and testing with a different one. Results show that the proposed model achieves an area under curve (AUC) as high as 0.97 with a selected set of features on the hold-out test set and 93% F1-score with a threshold=0.34.

Shukla et al. [36] propose a multi-layer condition-based model to identify fraudulent bots in Twitter. The proposed method analyzes Twitter-specific user profiles considering profile name, description, location, listed count, verification status, that are recovered with the Twitter API, and also

compare the string-based features with a bag of words containing tokens that are likely to be prevalent in bots. Authors performed a comparative evaluation of the proposed model with existing benchmark classifiers, such as SVN, Categorical Naïve Bayes, Bernoulli Naïve Bayes, Multilayer Perceptron, Decision Trees, Random Forest, and other automation identifiers, as well as with publicly available standard automation identifier API, such as Botometer, BotSentinel, and TweetBotOrNot. The Kaggle dataset is used for the experimentation, obtaining an accuracy of 99.0049% and a testing AUC of 0.996606.

Verma et al. [37] propose UCred, a model aimed at classifying user accounts as fake or real. UCred model separates text features of the user profile (e.g., description) from numeric features (e.g., followers, friends, verified flag) to be evaluated simultaneously by several machine learning and deep learning methods. Text features are evaluated by post-trained model (i.e., CNN, LSTM, BiLSTM) and pre-trained methods (i.e., BERT, RoBERT, DistilBERT), while numeric features are evaluated with Naïve Bayes, SVN, Random Forest, and K-Nearest Neighbor (KNN). Outputs from all these methods are submitted to a voting classifier to finally decide if the account is real or fake. To evaluate the performance, authors performed experiments on OSN dataset, which contains 1337 fake and 1481 real profiles. Results show that UCred reaches up to 98,96% of accuracy and 98,37% of F1-score.

Currently, there exist some automatic tools for bot detection in tweets in English, such as Debot [38], TweetBotOrNot [39], Bot Sentinel [40], and Botometer<sup>2</sup> [41].

Debot [38] is a system developed at the University of New Mexico which performs a search for correlated accounts by making use of deformed correlation (correlation extended by dynamic time warping) when analyzing their activity, given the idea that humans cannot be highly synchronous over long periods of time. The approach used in this system corresponds to an unsupervised method, not needing a labeled dataset to train a model beforehand. Debot works as a near real-time system, and has an accuracy of 94%, in addition to having an API that allows using it in external projects.

TweetBotOrNot [39] is an R package developed by Michael W. Kearney in 2018. It is based on the gradient boosted supervised model to obtain the probability of a Twitter account being a bot. To do so, the model uses user-level metadata, tweet-level, and tweet text, including friends, followers, posts, account time creation, retweets. The classifier was trained on 10,000 Twitter accounts with over 7000 bots, from which TweetBotOrNot shows an accuracy of 93.8%.

Bot Sentinel [40] is another Twitter analytics free tool founded in 2018 by Christopher Bouzy. Based on machine learning and artificial intelligence, this tool is able to identify and track trollbots and untrustworthy Twitter accounts. There is not much information available related to the

implementation details of this tool, however it has been used recently by other researchers in this domain.

Botometer [41] is a public tool designed by a joint project of the Observatory on Social Media and the Network Science Institute at Indiana University. Botometer is a supervised machine learning classifier that distinguishes bot-like and human-like accounts considering over 1000 features categorized into six classes: user profile, friends, network, temporal, content and language, and sentiment. The tool evaluates a Twitter account's activity and assigns it a score. Greater scores indicate increased bot-like activities. This service requires Twitter authentication and authorization in order to use. Both Botometer and Twitter API have rate limits, meaning that users can only make a certain number of queries in a given time period. Botometer has a website and API endpoints with similar functionality. Using the Botometer API requires keys associated with a Twitter app, when querying the API, users are responsible to send the required data (i.e., 200 most recent tweets by the account being checked and tweets mentioning this account) in a specified format through HTTPS requests.

All these works demonstrate the interest and feasibility of bot detection on Twitter. While most of these works report good results after testing, they do not fully match our requirements, regarding the consideration of English and Spanish and a prediction performance in real time, for the development of the functionality in our previous credibility model. Additionally, the datasets used are not available for general use; thus, a separate or further retraining is not possible. In the same way, although Alarifi et al. [32] perform supervised training at the end of their data collection process and design and implement a browser plug-in, none of these resources are available or enabled for use or incorporation into another project. Debot, TweetBotOrNot, and Bot Sentinel are still working today, but the teams in charge of those projects do not provide more keys for the use of their API; while Botometer has the same limitations of use as the Twitter API. Table 1 summarizes the points discussed, considering also the number of attributes used by each work as a reference.

The interest of this work is to be able to detect whether the user of the analyzed Twitter account is a bot or not, leaving the possibility of using additional analysis to subsequently infer whether the bot in question is a useful or dangerous bot.

## B. CREDIBILITY ANALYSIS CONSIDERING THE ACCOUNTS/USERS

Concerning on how to do the credibility analysis, the existing works consider the analysis of some types of information to calculate different credibility measures [26], [42], [43]. This information mainly includes four dimensions or levels associated to text, user or account, topic, and social influence. Text dimension measures the level of relevance and accuracy of the text, independent of the referenced topic or with respect to a certain topic. It is mainly calculated through

<sup>2</sup><https://botometer.osome.iu.edu/>



**TABLE 1. Twitter bot detection works.**

| Work                        | Real time detection | Language dependence | Learning method                                                                                      | API availability | Number of features for training | Reported results                 |
|-----------------------------|---------------------|---------------------|------------------------------------------------------------------------------------------------------|------------------|---------------------------------|----------------------------------|
| Chu et al., 2012 [17]       | No                  | Yes                 | Random Forest                                                                                        | No               | 8                               | Accuracy 96%                     |
| Alarifi et al., 2016 [33]   | Yes                 | Yes                 | Decision Tree; Bayesian network; SVM, Multilayer Artificial Neural Network                           | No               | 8                               | Accuracy 96%; Kappa coef. 0.61   |
| Kudugunta et al., 2018 [34] | No                  | Yes                 | LSTM                                                                                                 | No               | 16                              | Accuracy 96%                     |
| Barbon et al., 2018 [35]    | No                  | Yes                 | Random Forest                                                                                        | No               | 7                               | Accuracy 94,47%                  |
| Hayawi et al., 2022 [36]    | Yes                 | Yes                 | LSTM                                                                                                 | No               | 26                              | AUC 0,97%; F1-score 93%          |
| Shukla et al., 2022 [37]    | No                  | No                  | NLP; BoW                                                                                             | No               | 23                              | Accuracy 99,0049%; AUC 0.996606  |
| Verma et al., 2022 [38]     | No                  | Yes                 | CNN; LSTM; BiLSTM; BERT; RoBERT; DistilBERT; Naïve Bayes; SVN; Random Forest; KNN; voting classifier | No               | 17                              | Accuracy 98.96%; F1-score 0.9837 |
| Debot, 2016 [39]            | Yes                 | No                  | Clustering                                                                                           | Yes              | 7                               | Accuracy 94%                     |
| TweetBotOrNot, 2018 [40]    | Yes                 | Yes                 | Gradient Boosted                                                                                     | Yes              | Not available                   | Accuracy 93,8%                   |
| Bot Sentinel [41]           | Yes                 | Yes                 | Not available                                                                                        | Yes              | Not available                   | Not available                    |
| Botometer, 2022 [42]        | Yes                 | Yes                 | Random Forest                                                                                        | Yes              | 1000                            | AUC 0.99                         |

text analysis techniques. User dimension calculates the user account credibility based on attributes that describe it (e.g., the account creation date, if the account is verified, user gender and age). Social dimension calculates the credibility of a publication, related or not to a topic, based on the available metadata that describe the social impact of the user account and the post itself, with respect to other users. The social data, such as number of followers and friends, retweets, likes, are commonly used. Topic-level dimension measures the level of acceptance of the topic or event referenced in the text. It consists of identifying if the text refers to a specific topic or not, usually through text and sentiment analysis and topic detection techniques.

In the next, we survey some studies that have considered the user level for the credibility analysis. Al-Khalifa and Al-Eidan [44] developed a system to measure the credibility of arabic news content published in Twitter. The system uses two approaches to assign credibility levels (low, high, and average) to each tweet. The first approach is based on the similarity between Twitter posts (tweets) and authentic (i.e., verified) news sources. The second approach is based on the similarity with verified news sources in addition to a set of proposed features. The Twitter user profile features include username, user picture, real name, and verified account.

Namihira et al. [45] propose an expertise score by taking into account the user's knowledge (expertise). Expertise is the ratio of the number of tweets determined to be the same topic as that of the suspected tweet to the number of the past tweets of the target user. If the expertise score of the user

becomes higher, then tweets of the user are treated as a more reliable opinion even if it is a minority opinion. Likewise, if an author of the tweet refers to the same topics in his other tweets in the past, his/her expert score is increased. Afterward, the credibility is calculated using the expertise score for each user.

Alrubaian et al. [46] propose a real-time content credibility assessment system named CredFinder to measure the trustworthiness of information through user analysis and content analysis. The tweet and its metadata are sent to the extraction features process to generate a feature vectors. Afterward, all these data are used as input to the credibility score calculation algorithm. Data include the tweet content and the additional fields, such as the time of posting, the author name, number of followers, number of friends, hashtags, or mentions.

Hamdi et al. [47] propose an hybrid approach to evaluate information sources in term of credibility in Twitter. This approach relies on node2vec to extract features from Twitter followers/friends graph and also incorporates user features provided by Twitter. Authors analyze user-based features established by them as useful information for source of fake news detection. Such features represent the profile characteristics of users who have interacted with news on social media. The extracted features concerning the users are: created at, name, status count, default profile, and default profile image.

Iftene et al. [48] trained a neural network model using a collection of tweets that were manually annotated by human users as credible and not credible. For each individual tweet,

**TABLE 2.** Credibility works considering characteristics of users.

| Work                                | Credibility Levels               |                                                                      |                                  |       | Real-time       | Context                                           |
|-------------------------------------|----------------------------------|----------------------------------------------------------------------|----------------------------------|-------|-----------------|---------------------------------------------------|
|                                     | Text                             | User                                                                 | Social                           | Topic |                 |                                                   |
| Al-Khalifa & Al-Eidan, 2011 [45]    | Content                          | Username, user picture, real name, Account verification              | followers, following             | -     | Yes             | Credibility                                       |
| Namihira et al., 2013 [46]          | -                                | User expertise                                                       | -                                | Yes   | No              | Sentiment Analysis<br>Credibility                 |
| Alrubaihan et al., 2016 [47]        | Content, source<br>Time of tweet | Creation date, real name, username                                   | followers, following, retweets   | -     | Yes             | Credibility                                       |
| Hamdi et al., 2020 [48]             | -                                | Creation date, name status count, account and image profile          | followers/following social graph | Yes   | No              | SOFNs detection<br>Credibility                    |
| Iftene et al., 2020 [49]            | Content, URLs<br>Date of post    | Account verification, creation date location, description            | followers, retweets, favorites   | -     | Yes             | Fake users and news identification<br>Credibility |
| Hernandez-Mendoza et al., 2022 [50] | Content                          | Account verification, creation date                                  | following, followers retweets    | Yes   | Yes             | Credibility                                       |
| Saputri et al., 2023 [51]           | Content                          | Account verification, creation date, username, description, location | followers                        | -     | Yes<br>#covid19 | Credibility<br>fake news                          |

authors collect the following types of information concerning the user profile features: `hasLocation` (if user filled the location field), `hasDescription` (if user filled the description field), `hasGeo` (if user turned on geolocation), `isVerified` (if user was verified by Twitter), and `creationDate` (the date when account was created). Hernández-Mendoza et al. [49] extend a model originally proposed by [26] to analyze the credibility of social information sources, which was instantiated for Twitter. In this instantiated model authors calculate the credibility of a user considering whether the account is verified or not and the activity time of the account since its creation.

Saputri et al. [50] propose a method based on information gain and KNN to detect fake news related to COVID-19. The model is trained with a dataset built by authors and manually labeled as credible or not credible. Using the Twitter API, authors collected tweet with the hashtag `#covid19`, along with the users' profile, consisting on username, location, user description, account creation time, number of followers, as well as user verification, from which the author's credibility is determined and the manual labeling is done. From the text of tweets, the information gain is computed based on the TF-IDF method. Thus, from the user's credibility and the information gain, the KNN is able to identify tweets related to COVID-19 as fake news or valid information.

Table 2 summarizes these works considering the different credibility levels (text, user, social, and topic), if the work can be applied in realtime or not, and the application context in which they were developed. These studies demonstrate that the credibility of the user of Twitter account is relevant for credibility analysis. In this work, besides considering the common features of users to calculate the user credibility

level, we enrich the model by detecting the nature of the user (bot or not bot).

### III. EVALUATION OF METHODS FOR BOT DETECTION IN TWITTER

For the evaluation of algorithms for bot detection, we follow the methodology shown in Figure 1, which includes steps from data collection and cleaning, to training of the predictive models and their evaluation. This methodology is inspired on the classical methodology for Knowledge Discovery in Databases (KDD), which proposes five phases for data processing (i.e., data selection, preprocessing, transformation, data mining, and interpretation/evaluation). In this work, we correspond these phases such as database configuration (data selection), preprocessing (preprocessing and transformation), training phase (data mining), and testing and analysis (interpretation/evaluation), as Figure 1 illustrates. In the following, we detail each phase.

#### A. DATASET CONFIGURATION

The dataset configuration consists of two steps: feature engineering and data extraction from Twitter, as we explain in the following.

##### 1) FEATURE ENGINEERING STEP

The first step of our methodology consists in making a feature engineering, whose result is used in the training of the predictive models. Table 2 shows works that use social features of the analyzed account [44], [46], [47], [48], [49], [50], others works consider features of the user itself [44], [45], [46], [47], [48], [49], [50], and others works use structural features associated with the text content [44], [46], [48], [49], [50]. These features have been also considered in

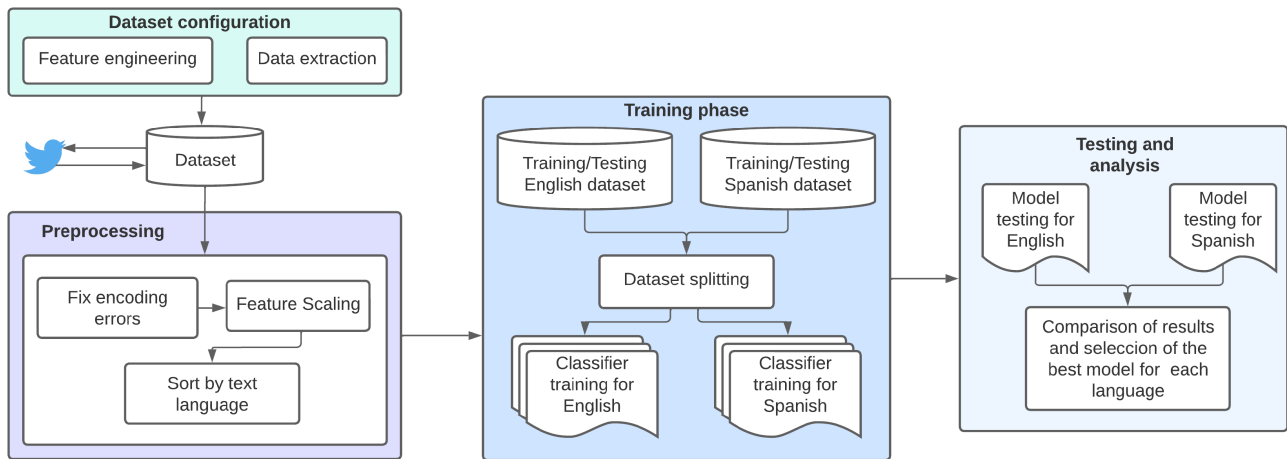


FIGURE 1. Methodology for training and validation of bot detection algorithms.

our credibility model. For the new extended credibility model which includes bot detection, some additional user features considering their relevance for the training of predictive models studied are selected [32], [33], [34]. Similar to features selected in the works presented by [32] and [33], in our work we also consider some social features of the user account. We further aggregate data on the text or content of the tweet, considering structural features as in [33]. We have also considered a new feature based on the Discrete Wavelet Transform (DWT) [34], which have been used for text analysis. The features selected and used in this work for the construction of the training datasets and subsequent use in the training process include:

- 1) **Followers\_count**: The number of followers the account has.
- 2) **Favorite\_count**: The total number of times a tweet has been favourite.
- 3) **Favourites\_count**: The number of tweets a user has liked.
- 4) **Listed\_count**: The total number of public lists in which the user is a member.
- 5) **Friends\_count**: The total number of users the user has added as friends.
- 6) **Retweet\_count**: The total number of retweets of the user.
- 7) **Statuses\_count**: The total number of tweets the user has or has made on their accounts.
- 8) **Num\_URLs**: The number of URLs contained in the tweet to analyze.
- 9) **Num\_hashtags**: The number of hashtags contained in the tweet to analyze.
- 10) **Num\_mentions**: The number of mentions contained in the tweet to analyze.
- 11) **Wavelet average**: A centroid calculated from vectorization of the tweet text based on the DWT.
- 12) **URL\_ratio**: The ratio of the number of URL present in the text to the total number of words in the tweet.

- 13) **Mention\_ratio**: The ratio of the number of mentions present in the text to the total words in the tweet.
- 14) **Hashtag\_ratio**: The ratio of the number of hashtags present in the text to the total words in the tweet.
- 15) **Entropy**: Value of the Shannon entropy formula.

The list of features corresponds to social features or quantitative information from the text itself. Most features can be retrieved from the Twitter account itself using the Twitter API. The features 3) Favourites\_count, 6) Retweet\_count, 8) Num\_URLs, 9) Num\_hashtags, and 10) Num\_mentions were also used by Kudugunta et al. [33], while features 1) Followers\_count, 2) Favorite\_count, 4) Listed\_count, and 5) Friends\_count were used by Alarifi et al. [32]. The feature 11) Wavelet average considers the DWT to obtain spectral information according to the content of the tweet text, such as was proposed by Barbon et al. [34]. The text of the tweet is vectorized using the spaCy library, differentiating whether it is a text in Spanish or English, and then calculating the DWT of the vector. Finally, an average of the complete vector is performed and this result is used as a feature for classification. Originally, authors in [34] use the DWT to obtain spectral information according to the content of the tweet text and to characterize the writing style embedded in post contents. For the proposal of this particular work, a similar technique is used, in the sense that the content of the tweet is transformed into a vector to which the DWT is calculated and then its centroid is computed. The centroid was not considered in the original procedure proposed by Barbon et al. [34].

Features 12) URL\_ratio, 13) Mention\_ratio, and 14) Hashtag\_ratio are structural text features based on ratios that are considered interesting to study. Some ratios provide additional information which can be useful. For example, Kudugunta et al. [33] use a feature that considers the ratio between followers and friends (following) of the account. The idea is to be able to see if the ratio between the number of words composing a tweet and the number of mentions/URLs/hashtags could be an important feature that

helps to define a pattern of behavior of social bots on Twitter. Feature 15) Entropy is based on the Shannon's Entropy calculation. This feature measures the amount of information contained in the text of the tweet, lower values means less information the text contains. The entropy has been also used in similar contexts of bots detection [7], [51], [52].

## 2) DATA EXTRACTION

Although there are a variety of works that offer the possibility of requesting the datasets used for their studies based on Twitter, most of them only have the id of the labeled accounts and not their respective tweets. Moreover, we found that some of these datasets are obsolete with deleted accounts or they are now totally different from the period in which they were collected [53]. Other authors have created their own datasets using the Twitter API, but they are not reported as public [54]. The task of making a full dataset containing tweets and data related to them demands many resources, such as the labeling process and the data gathering from Twitter, where the API offers limitations for free accounts. For instance, authors in [55] report a time of four months to collect data from 1800 Twitter accounts.

Other studies, such as the one presented in [56], focus on the prediction models and use public datasets proposed by other works [28], [57].

In this work, we requested and used the datasets collected by authors in [28]. These datasets contain different user accounts and their tweets in a variety of languages, such as English, Spanish, French, Japanese. No additional requests to the Twitter API were required. Table 3 shows the total number of user accounts contained in the different datasets (genuine, social spambots, traditional spambots, and fake followers), the number of tweets associated to them, and the year which represents the average of the creation years of the accounts that belong to the dataset. This table is a simplified version of the original table presented in [28].

## B. PREPROCESSING

After obtaining the datasets (Table 3), an initial review showed that they contain several encoding errors (accents, punctual letters such as 'ñ' or capital letters), emojis, and characters that do not contribute to the content of the text of the tweet. Additionally, the text associated to tweets are in a variety of languages, such as English, Spanish, French, Japanese. For these reasons, the preprocessing to prepare the data for training involves data cleaning and data filter tasks, such as:

- **Fix encoding errors:** The Python library *fffy* is used to correct text encoding problems in tweets. This consideration is necessary mainly when preprocessing the information that would compose the training dataset in Spanish, since several characters had encoding problems, such as accents (á, é, í, ó, and ú), umlauts, or the letter 'ñ'.
- **Feature scaling:** Due to different feature scales, they are standardized by removing the mean and scaling

to unit variance. For this, features are scaled using the Scikit-learn library *StandardScaler*. This function calculates the standard score of a sample  $x$  as:  $z = (x - u)/s$ , where  $u$  is the mean and  $s$  is the standard deviation of the training samples.

- **Filter by text language:** A function is used to identify the language in which the tweet was written with the Python library *langid*; thus, it is possible to make a quick classification of texts written in English or Spanish tagging them as 'en' or 'es', respectively.

Once the preprocessing is finished, the relevant records are collected between the group of genuine (human) accounts and the rest of the bot groups (social spambots #1, #2, and #3, traditional spambots #1 and #2), obtaining two datasets. Table 4 shows the statistics associated to each dataset distributed by language and category (human or bot).

The total number of rows for the Spanish training dataset is limited by the number of tweets made by bots. Although there is a larger number of tweets in Spanish from genuine (human) users, we decide to take a similar number of tweets from bots in order to keep the training dataset as balanced as possible. Although more data are available for the English training dataset, we try to do not exceed by much the size of the Spanish dataset.

### B.1. Feature selection methods:

In the interest of reducing the high dimensionality of datasets and to provide more parsimonious models, different methods of feature selection have been studied and classified into three groups: filters, wrappers, and embedded [58]. However, no single selection algorithm into these groups seems to be capable of ensuring optimal results in terms of predictive performance, robustness, stability, and interpretability. In this sense, we have explored two feature selection methods which have been used successfully in different domains. The first one corresponds to an embedded method, feature importance in Random Forest [59], [60], [61], [62]. The second one considered is the SHapley Additive exPlanations (SHAP) method, which help understanding what are the main features that affect the output of the model [63], and is also being used as a feature selection mechanism [64], [65], [66]. In this context, the feature selection is assisted by SHAP – i.e., the features are selected and grouped according to SHAP values.

### 1) FEATURE IMPORTANCE IN RANDOM FOREST

As an interesting result derived from the importance of the features in the training process of the data with the Random Forest, some features with higher variance were involved in the classification process (i.e., *favourites\_count*, *statuses\_count*, *followers\_count*, *friends\_count*, and *listed\_count*). Figure 2 and Figure 3 show these best features. The results correspond to the permutation feature importance of the Random Forest.<sup>3</sup> This is a model inspection approach that can be used to any fitted estimator and is especially

<sup>3</sup>[https://scikit-learn.org/stable/auto\\_examples/ensemble/plot\\_forest\\_importances.html](https://scikit-learn.org/stable/auto_examples/ensemble/plot_forest_importances.html)



TABLE 3. Datasets description.

| Dataset                 | Description                                                             | Accounts | Tweets    | Year |
|-------------------------|-------------------------------------------------------------------------|----------|-----------|------|
| Genuine accounts        | verified accounts that are human-operated                               | 3.474    | 8.377.522 | 2011 |
| Social spambots #1      | retweeters of an Italian political candidate                            | 991      | 1.048.575 | 2012 |
| Social spambots #2      | spammers of paid apps for mobile devices                                | 3.457    | 428.542   | 2014 |
| Social spambots #3      | spammers of products on sale at Amazon.com                              | 464      | 1.048.575 | 2011 |
| Traditional spambots #1 | training set of spammers                                                | 1.000    | 145.094   | 2009 |
| Traditional spambots #2 | spammers of scam URLs                                                   | 100      | 74.957    | 2014 |
| Traditional spambots #3 | automated accounts spamming job offers                                  | 433      | 5.794.931 | 2013 |
| Traditional spambots #4 | another group of automated accounts spamming job offers                 | 1.128    | 133.311   | 2009 |
| Fake followers          | simple accounts that inflate the number of followers of another account | 3.351    | 196.027   | 2012 |

TABLE 4. Training dataset in spanish and english.

|                        | Datasets |         |
|------------------------|----------|---------|
|                        | Spanish  | English |
| Data quantity (tweets) | 12557    | 25024   |
| Features               | 15       | 15      |
| Data tagged as "bot"   | 5231     | 12852   |
| Data tagged as "human" | 7326     | 12172   |

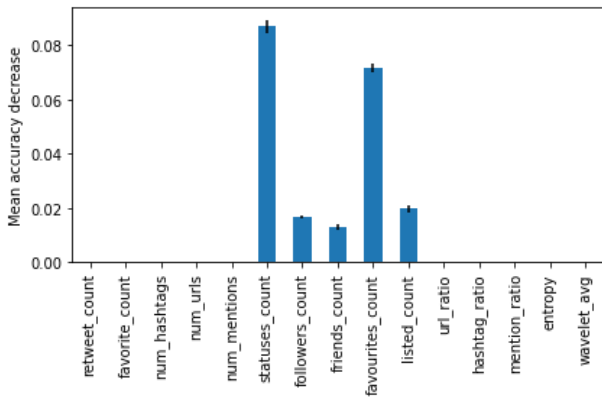


FIGURE 2. Feature importance using permutation on full model (English).

helpful for estimators that are non-linear or opaque. The permutation feature significance is defined as the reduction in model score caused by randomly rearranging a single feature value. The technique destroys the link between the feature and the goal, and the decrease in model score reflects how much the feature is relied upon. This method has the advantage of being model independent and can be calculated numerous times with various permutations of the characteristic [67]. The disadvantages of impurity-based feature importance are overcome by permutation feature importance since it does not favor high-cardinality features and may be computed on a test set with missing data.

## 2) SHAP METHOD

SHAP (SHapley Additive exPlanations), a unified framework for interpreting predictions, assigns each feature an importance value for a particular prediction. This method has also been adopted to provide a reasonable interpretable model [63]. The SHAP value method has been used in a number of cases to calculate the importance of features

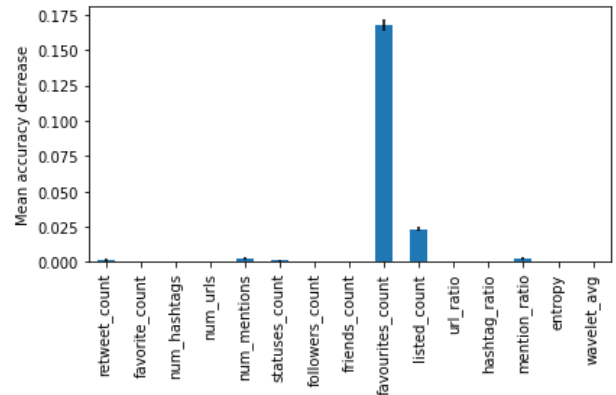


FIGURE 3. Feature importance using permutation on full model (Spanish).

in machine learning models, so as to get an intuitive explanation [68]. Figure 4 and Figure 5 show the SHAP values on our two datasets.

In furtherance of testing the impact of applying a feature selection method, we have reduced the number of features to 7 and 5 for Spanish and English datasets, respectively. Afterwards, we have retrained our classifiers, resulting Random Forest (as classifier) with the best metrics. Selected features correspond to an importance score greater than 1 in Table 5. Table 6 shows metrics resulting from the prediction on the testing sets using Random Forest. We can observe that the metrics are good with values over 90% and the selected features are associated to social features. We could infer from these results that these features are independent of the language and the tweet content (note that the associated features to language and content were not selected by the feature selection methods). However, we prefer to remain conservative about this insight since more cases such as other languages have to be analyzed, thus we have trained our classifiers with the whole set of features – i.e., fifteen features (Section III-A1). We would like to explore, as a future work, other languages different to Spanish and English, to measure the real impact of the language and the tweet content in the bot detection; which also demands to include new datasets.

## C. TRAINING AND SELECTION OF PREDICTIVE MODELS

Since the writing style in English and Spanish has different structures that could hinder the analysis, we consider two

TABLE 5. SHAP feature values.

| Feature number | Feature name (Spanish) | Importance | Feature name (English) | Importance |
|----------------|------------------------|------------|------------------------|------------|
| 0              | favourites_count       | 30.449375  | favourites_count       | 18.714688  |
| 1              | statuses_count         | 4.948437   | statuses_count         | 14.826875  |
| 2              | followers_count        | 4.694687   | listed_count           | 8.447188   |
| 3              | friends_count          | 2.247812   | followers_count        | 7.021250   |
| 4              | listed_count           | 1.743125   | friends_count          | 1.870937   |
| 5              | num_mentions           | 1.186562   | retweet_count          | 0.355937   |
| 6              | mention_ratio          | 1.128125   | url_ratio              | 0.248437   |
| 7              | retweet_count          | 0.239375   | entropy                | 0.199375   |
| 8              | url_ratio              | 0.192188   | num_mentions           | 0.180938   |
| 9              | entropy                | 0.097187   | wavelet_avg            | 0.180000   |
| 10             | num_urls               | 0.091250   | mention_ratio          | 0.169687   |
| 11             | wavelet_avg            | 0.063438   | num_urls               | 0.147812   |
| 12             | hashtag_ratio          | 0.053125   | favorite_count         | 0.138750   |
| 13             | num_hashtags           | 0.035937   | hashtag_ratio          | 0.090312   |
| 14             | favorite_count         | 0.029375   | num_hashtags           | 0.080937   |

TABLE 6. Best testing results using features selected with SHAP for spanish and english.

|         | Classifier    | Number of Features | Precision | Recall | F1 score | Accuracy | AUC Score |
|---------|---------------|--------------------|-----------|--------|----------|----------|-----------|
| Spanish | Random Forest | 7                  | 0.9133    | 1.0000 | 0.9547   | 0.9450   | 0.9347    |
| English | Random Forest | 5                  | 0.9575    | 0.9904 | 0.9737   | 0.9732   | 0.9731    |

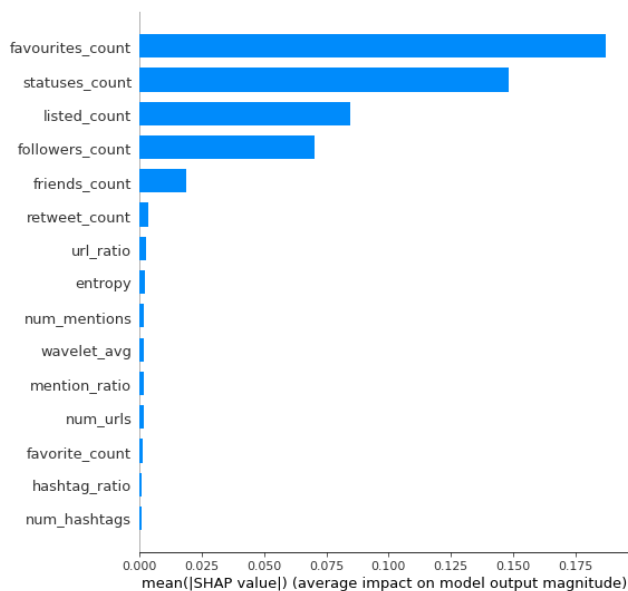


FIGURE 4. SHAP values for english dataset.

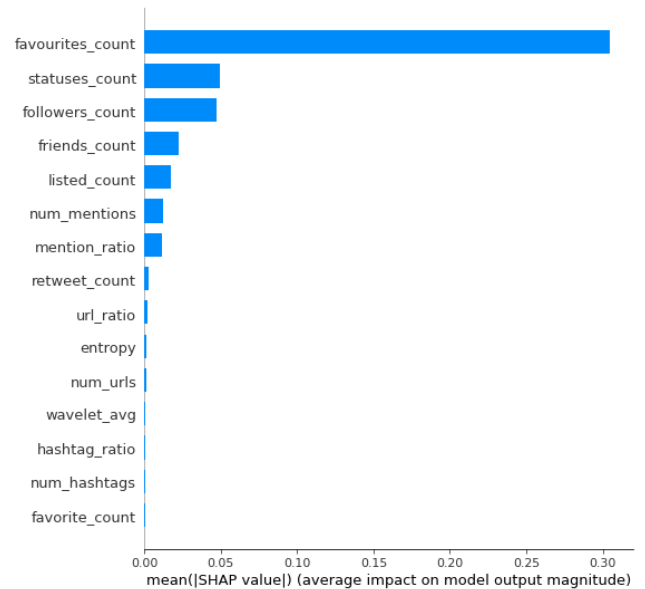


FIGURE 5. SHAP values for spanish dataset.

separated models for training and classification, instead of one general predictive model. In the following, we present the experimental conditions and training details.

### 1) THE EXPERIMENTAL CONDITIONS

In this work, we deal with a supervised learning problem with a binary classification task considering two classes: “bot” and “human”. In this context, there exist several machine learning algorithms that have shown good results for bot detection [69], [70]. We have considered some of the most popular and relevant:

- **AdaBoost classifier:** It is a meta-estimator that starts by fitting a classifier to the initial dataset and then fits

additional copies of the classifier on the same dataset with the weights of instances that were mistakenly categorized. This is done such that future classifiers concentrate more on challenging situations [71], [72].

- **Bagging classifier:** It is an ensemble meta-estimator that applies basic classifiers to different random subsets of the initial dataset, and then combines each prediction (either by voting or by averaging) to get the final prediction. By adding randomization to the process of building a black-box estimator (such a decision tree), a meta-estimator of this kind can often be used to lower the variance of the estimator [73].
- **Decision Tree classifier:** It is a supervised learning technique that is non-parametric and is commonly

used in classification tasks. The objective is to learn straightforward decision rules derived from the data features in order to build a model that predicts the value of a target variable. It is a technique capable of performing multi-class classification on a dataset [74].

- **Logistic Regression:** Instead of a model used in a regression task, this one is used for classification. In the literature, logic regression, maximum-entropy classification (MaxEnt), and the log-linear classifier are also used to refer to logistic regression. In these models a logistic function is used to simulate the probabilities describing the potential outcomes of a single experiment [75].
- **Random Forest classifier:** It is a meta estimator that employs averaging to increase predicted accuracy and reduce overfitting after fitting numerous decision tree classifiers to different dataset subsamples [67].
- **K-Neighbors classifier (KN):** Neighbors-based classification is a type of instance-based learning or non-generalizing learning. The nearest neighbor decision rule assigns to an unclassified sample point the classification of the nearest of a set of previously classified points. Thus, from a simple majority vote of the nearest neighbors of each point a classification is computed. A query point is assigned the data class which has the most representatives within the nearest neighbors of the point [76].
- **Support vector machine (SVM):** SVM is a machine learning method widely used for solving classification and regression problems. The main idea of the SVM algorithm is to build an optimal hyperplane with the largest margin that best separates the data into classes [77]. The SVM is a popular algorithm due to its high generalization capability together with its ability to find global and non-linear classification solutions. In this work, we use Support Vector Classification (SVC), the scikit-learn implementation of this method for classification purposes.
- **Multi-layer Perceptron (MLP):** MLP is a supervised learning algorithm that learns a function by training on a dataset. Given a set of features or input of  $m$  dimensions and a target of  $o$  dimension, it can learn a non-linear function approximator for either classification or regression [78].

To determine the best parameters, each algorithm was trained several times by exploring different parameter values using a GridSearchCV method.<sup>4</sup> This method exhaustively considers all parameter combinations over specified parameter values for each classifier. The parameters of the classifier used to apply these methods are optimized by cross-validated grid-search over a parameter grid. Table 7 summarizes the parameters used for each algorithm.

The classification metrics<sup>5</sup> used to quantify the predictions quality of the algorithms selected are precision, recall, F1-score, accuracy, and AUC score. Considering the number of true positives (tp), false positives (fp), false negatives (fn), and true negatives (tn), these metrics reach its best value at 1 and worst score at 0, calculated as follows:

- **Precision:** It is the ratio  $\frac{tp}{tp+fp}$ . The precision is intuitively the ability of the classifier not to label as positive a sample that is negative.
- **Recall:** It is the ratio  $\frac{tp}{tp+fn}$ . The recall is intuitively the ability of the classifier to find all the positive samples.
- **F1-score:** It can be interpreted as a harmonic mean of the precision and recall. The relative contribution of precision and recall to the F1-score is equal, calculated as  $2 * \frac{precision * recall}{precision + recall}$ .
- **Accuracy:** This function computes the proportion of correct classifications, as  $\frac{tp+tn}{tp+fn+tn+fp}$ .
- **AUC score:** Compute Area Under the Receiver Operating Characteristic Curve (ROC AUC) from prediction scores. It is created by plotting the fraction of  $tp$  out of the positives vs. the fraction of  $fp$  out of the negatives, at various threshold settings [79].

## 2) TRAINING

We proceeded to train with different machine learning algorithms to analyze which of them would be the most suitable to use as classifier for bot detection. We use the library scikit-learn<sup>6</sup> in Python to create a script to train, test, and visualize the results obtained by each algorithm. For each language, datasets are divided by 80% for training and 20% for testing. The *training set* is used in the cross-validation process, while the *test set* is subsequently used in the validation phase.

Due to the cross-validation process, each algorithm is performed four times (i.e.,  $cv = 4$ , the parameter for the cross-validation) with the training set. This cross-validation returns stratified folds – i.e., the folds are made by preserving the percentage of samples for each class. The hyperparameter space is visited to search for the best cross validation score. In this way any parameter provided when constructing a classifier is being optimized. Table 8 shows the GridSearchCV results. All the possible combinations of parameter values are evaluated and the best combination is retained in terms of accuracy score. The mean and standard deviation of test scores are reported.

## D. TESTS AND RESULTS ANALYSIS

In this section, we present the results obtained in the testing phase of the selected models once they were trained. Two kinds of validation are performed. The internal validation is performed with the 20% of each dataset from the splitting phase and which purpose is testing. The external validation

<sup>4</sup>[https://scikit-learn.org/stable/modules/generated/sklearn.model\\_selection.GridSearchCV.html](https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.GridSearchCV.html)

<sup>5</sup>[https://scikit-learn.org/stable/modules/model\\_evaluation.html#classification-metrics](https://scikit-learn.org/stable/modules/model_evaluation.html#classification-metrics)

<sup>6</sup><https://scikit-learn.org/stable/>

**TABLE 7.** Hyperparameters used in the gridsearch for each algorithm.

| Classifier          | Hyperparameter array                                                                                                                                                                                        |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adaboost            | base_estimator = [Decision Tree, SVC, GaussianNB]; n_estimators = [10, 50, 80, 120]; learning_rate = [1.0, 5, 6.5, 8]; algorithm = [SAMME, SAMME.R]; random_state = [1,5,7]                                 |
| Bagging             | base_estimator = [Decision Tree, SVC, GaussianNB]; n_estimators = [10, 25, 30, 50]; max_samples = [1.0, 2.5, 5, 10]                                                                                         |
| Decision Tree       | criterion = [gini, entropy]; splitter = [best, random]; max_depth = [100, 120, 150, 200]; min_samples_split = [2, 5, 1.5, 3.5]; min_samples_leaf = [1, 1.6, 3, 4.2]; max_features = [auto, sqrt, log2]      |
| K-Neighbors         | n_estimators = [5, 10, 15, 20]; weights = [uniform, distance]; algorithm = [auto, brute, kd_tree, ball_tree]; leaf_size = [15, 30, 45, 60]; metric = [minkowski]                                            |
| Logistic Regression | penalty = [l1, l2, elasticnet]; dual = [True]; C = [1.5, 2.0, 3.8]; solver = [lbfgs, liblinear, cholesky]                                                                                                   |
| MLP                 | hidden_layer_sizes = [(5,2), (8,3), (11,4)]; activation = [relu, identity, tanh, logistic]; max_iter = [1000, 1200]; solver = [adam, lbfgs, sgd]; random_state = [2, 5, 8]; apha = [0.0001, 0.0010, 0.0100] |
| Random Forest       | criterion = [gini, entropy]; max_depth = [50, 80, 90, 100, 110, 150, 180, 200]; min_samples_split = [10, 12, 15, 20]; min_samples_leaf = [2, 8, 11, 15]; n_estimators = [100, 300, 500, 1000]               |
| SVC                 | C = [1.0, 1.5, 2.5]; kernel = [poly, linear, rbf]; degree = [2, 3, 5]; gamma = [scale, auto]; decision_function_shape = [ovo, ovr]                                                                          |

**TABLE 8.** Parameters of the best results in spanish and english datasets.

| Classifier          | Parameters                                                                                                            | Mean score | STD score |
|---------------------|-----------------------------------------------------------------------------------------------------------------------|------------|-----------|
| Adaboost            | base_estimators = Decision Tree Classifier; n_estimators = 10; learning_rate = 8; algorithm = SAMME; random_state = 5 | 0.99871    | 0.00033   |
| Bagging             | base_estimators = Decision Tree Classifier; n_estimators = 50; max_samples = 1                                        | 0.99900    | 0.00045   |
| Decision Tree       | criterion = gini; splitter = best; max_depth = 200; min_samples_split = 2; min_samples_leaf = 1; max_features = auto  | 0.99721    | 0.00096   |
| Logistic Regression | penalty = l2; C = 2; solver = liblinear, dual = True                                                                  | 0.94883    | 0.00214   |
| MLP                 | activation = relu; alpha = 0.0001; hidden_layer_sizes = (5,2), max_iter = 1000, random_state = 8; solver = lbfgs      | 0.99662    | 0.00091   |
| K-Neighbors         | algorithm = ball_tree; leaf_size = 15; metric = minkowski, n_neighbors = 5, weights = distance                        | 0.95869    | 0.00198   |
| SVC                 | C = 2.5; decision_function_shape = ovo; degree = 2, gamma = scale, kernel = rbf;                                      | 0.97133    | 0.00213   |
| Random Forest       | n_estimators = 100; criterion = gini; max_depth = 180; min_samples_split = 20; min_samples_leaf = 2                   | 0.99811    | 0.00043   |
| Adaboost            | base_estimator = Decision Tree Classifier; n_estimators = 10; learning_rate = 1; algorithm = SAMME; random_state = 5  | 0.99935    | 0.00057   |
| Bagging             | base_estimator = Decision Tree ; n_estimators = 25; max_samples = 1; bootstrap = False; bootstrap_features = True     | 0.99965    | 0.00009   |
| Decision Tree       | criterion = gini; splitter = best; max_depth = 120, min_samples_split = 2, min_samples_leaf = 1; max_features = sqrt  | 0.99535    | 0.00222   |
| Logistic Regression | penalty=l2; C = 3.8; solver = liblinear; dual = True                                                                  | 0.92212    | 0.00319   |
| MLP                 | activation= tanh; alpha = 0.0001, hidden_layer_sizes = (8, 3); max_iter = 1200; random_state = 2; solver = lbfgs      | 0.99471    | 0.00022   |
| K-Neighbors         | algorithm= auto; leaf_size = 15, metric = minkowski; n_neighbors = 5; weights = distance                              | 0.94305    | 0.00355   |
| SVC                 | C= 2.5; function_shape = ovo, degree = 2; gamma = gamma; kernel= rbf                                                  | 0.94690    | 0.00354   |
| Random Forest       | n_estimators = 100; criterion = entropy; max_depth = 180; min_samples_split = 12; min_samples_leaf = 2                | 0.99925    | 0.00030   |

corresponds to test performed on new cases of external sources.

## 1) INTERNAL VALIDATION

Table 9 summarizes metrics resulting of the prediction with each best classifier (Table 7) fitted to the testing datasets (from the splitting of the training datasets). Likewise, Figure 6 and Figure 7 show the confusion matrices corresponding to these classifiers.

In terms of precision and accuracy in the validation phase, the worst values are for Bagging with 55.76% and 52.98%, respectively for the Spanish dataset, and Decision tree with 64.79% and 72.42% for the English dataset. While the rest of algorithms got metrics over 93% for the same metrics on both datasets (except for Adaboost with 56.39% and 54.10% for the Spanish dataset). The best results were Random Forest and SVC for Spanish, and Random Forest and MLP for English dataset.

In order to use the best classifier for the following tests, and include a model in further implementations, we select Random Forest. This selection corresponds to the capability to detect bots (true negative values in the confusion matrices), and the possibility of incorporate more explicability of our models [80]. Additionally, Random Forest only learns from

a small subset of features and the random feature selection makes the trees more independent of one another than other classifiers, which frequently yields better predictive performance (due to better variance-bias trade-offs). This makes that random feature selection be the process faster than bagging [81]. The “bagging” methods, such as Bagging and Random Forest are designed to simplify models that overfit the training data.

## 2) EXTERNAL VALIDATION

In the interest of analyzing of the effectiveness of our classifiers based on Random Forest, we randomly selected 10 accounts, both in English and Spanish, and classified them as bot or human, using the Botometer program and our trained models. Table 10 and Table 11 show the results obtained. The Prediction columns correspond to the result obtained by the our models. Results of Botometer are colored darker as they are closer to 5 to indicate that the account is more likely to be an automated account (or bot); in contrast, the closer it is to 0, the more likely it is a genuine person. Botometer uses a number of characteristics associated to Twitter accounts to generate these results – e.g., volume of political tweets the accounts share (retweet), how many posts they make over time. Although we cannot be sure that the values closest to



**TABLE 9.** Classifier results for spanish and english dataset.

|         | Classifier          | Prediction |        |          |          |           |
|---------|---------------------|------------|--------|----------|----------|-----------|
|         |                     | Precision  | Recall | F1-score | Accuracy | AUC Score |
| Spanish | Decision tree       | 0.9761     | 0.8735 | 0.9220   | 0.9144   | 0.9221    |
|         | Random Forest       | 0.9664     | 1.0000 | 0.9761   | 0.9717   | 0.9664    |
|         | Bagging             | 0.5576     | 0.9113 | 0.6918   | 0.5298   | 0.4580    |
|         | Adaboost            | 0.5639     | 0.9154 | 0.6979   | 0.5410   | 0.4705    |
|         | MLP                 | 0.9535     | 0.9725 | 0.9629   | 0.9566   | 0.9536    |
|         | K-Neighbors         | 0.9728     | 0.9353 | 0.9537   | 0.9474   | 0.9497    |
|         | SVC                 | 0.9920     | 0.9408 | 0.9657   | 0.9613   | 0.9652    |
|         | Logistic Regression | 0.9866     | 0.9127 | 0.9482   | 0.9422   | 0.7478    |
| English | Decision tree       | 0.6479     | 0.9844 | 0.7815   | 0.7242   | 0.7238    |
|         | Random Forest       | 0.9534     | 0.9888 | 0.9708   | 0.9702   | 0.9701    |
|         | Bagging             | 0.9325     | 0.9984 | 0.9643   | 0.9630   | 0.9629    |
|         | Adaboost            | 0.9475     | 0.9445 | 0.9460   | 0.9460   | 0.9460    |
|         | MLP                 | 0.9954     | 0.9549 | 0.9747   | 0.9752   | 0.9752    |
|         | K-Neighbor          | 0.9496     | 0.9258 | 0.9375   | 0.9382   | 0.9382    |
|         | SVC                 | 0.9497     | 0.9341 | 0.9418   | 0.9422   | 0.9422    |
|         | Logistic Regression | 0.9561     | 0.8791 | 0.9160   | 0.9192   | 0.9193    |

**FIGURE 6.** Confusion matrix for each algorithm in test (Spanish).**FIGURE 7.** Confusion matrix for each algorithm in test (English).**TABLE 10.** Spanish model test results.

| User             | Botometer | Prediction |
|------------------|-----------|------------|
| @_BeejournalisT_ | 0.6       | human      |
| @SpanishDict     | 4.3       | bot        |
| @chile_soberano  | 3.4       | human      |
| @Capitanleviack1 | 1.6       | human      |
| @soldelpilarmr   | 0.4       | human      |
| @artesano robert | 0.6       | human      |
| @CNNChile        | 3.2       | bot        |
| @memewordES      | 4.8       | bot        |
| @gastonoyarzun2  | 1.6       | human      |
| @Migonver        | 0.2       | human      |

**TABLE 11.** English model test results.

| User             | Botometer | Prediction |
|------------------|-----------|------------|
| @RicFlairNatrBoy | 1.2       | human      |
| @tommyinnit      | 1.3       | human      |
| @ABC7NY          | 2.9       | bot        |
| @Wario64         | 1.8       | human      |
| @introvertsmemes | 3.3       | bot        |
| @Carl_E_Miller   | 1.6       | human      |
| @RickyCa09039186 | 3.7       | bot        |
| @SportsCenter    | 1         | human      |
| @EmilesAsylum    | 1.6       | human      |
| @BlancoBraxx     | 1.4       | human      |

5 in the Botometer column really correspond to a bot, we can see that there is similar values of our models with respect to the Botometer scores.

Table 10 shows that, of the 4 values in the table close to 5 (@SpanishDict, @chile\_soberano, @CNNChile, and @memewordES), the Random Forest model for Spanish only identified three of them as bot. Analyzing the

@chile\_soberano account more specifically, it is a Twitter account that is dedicated to posting and sharing tweets that have political content, mostly of them jokes or satire. This indicates the high probability that the account in question is a hybrid account being managed by a genuine person and at the same time with some degree of automation, hence the classification as “human” made by our model. However, having politics as its main topic, Botometer may have considered it more likely as a bot that shares political tweets with a high volume. The rest of the results obtained for this test seems aligned with the results obtained with Botometer. Concerning the bot accounts, these have a bot behavior. @CNNChile is a news account which usually have a hybrid account behavior, since they usually schedule their publications. @SpanishDict is an account for learning Spanish words. The structure of their tweets is basically the same for each post, being almost identical the hashtags added to each publication. @mememewordES is a memes account that shows humorous images with their Spanish translation in the text of the post. Botometer labeled it as a bot in its repository and with a spam behavior. The structure of the tweets is always the same, with content taken from other pages.

Table 11 shows a comparison between the results obtained in Botometer and our Random Forest based model for English. Three accounts were considered as bots (@ABC7NY, @introvertsmemes, and @RickyCa09039186). @ABC7NY is a news account, usually tends to behave like a hybrid account, since they tend to schedule their posts. @introvertsmemes is a memes account that displays humorous images and texts related to the aspect of being introverted. While the structure of the posts is diverse, Botometer indicates that it has been found as a bot within their repositories, as well as possibly having been reported as a bot by user feedback. @RickyCa09039186 is an account that publishes text posts on predefined formatted topics. It is usually with political topics. Botometer indicates that it has been found in its repositories as a bot, spam behavior and possibly been reported as a bot by user feedback. In the case of the @tommyinit account, this is the account of a person who has a presence on the YouTube platform, his comments usually have images attached, with short texts that rarely have mentions, hashtags, or URLs, so that the model has considered his behavior as typical of a genuine user. The rest of the predictions made to the other accounts have a good fit with the probabilities calculated by Botometer.

Considering these tests, Random Forest based models for Spanish and for English, obtain good results when working with arbitrary Twitter data.

Following sections describe how the bot detection can be used for credibility analysis.

#### IV. A CREDIBILITY MODEL FOR TWITTER

A credibility model to analyze social media, based in three levels of credibility (i.e., text, user, and social impact) was proposed in a previous work [26]. This original credibility

model was instantiated in T-CREo, a credibility analysis framework for Twitter [27] and used to perform data source comparison [82], [83]. T-CREo works as a Google Chrome extension to perform an automatic credibility analysis using web scraping techniques and the Twitter API to achieve this analysis in real time. In this section, we describe the bases of the original credibility model.

##### A. TEXT CREDIBILITY

Text credibility analyzes syntactically the content of the post (without checking the author attributes), through SPAM, bad words, and misspelling filters, as shown in Definition 1.

**Definition 1: Text Credibility (TextCred).** Given the text of a post  $p$ , represented as  $p.text$ , Text Credibility is a function, denoted as  $TextCred(p.text)$ , that returns a measure  $\in [0, 100]$ , defined as:

$$\begin{aligned} TextCred(p.text) = & w_{SPAM} \times IsSPAM(p.text) \\ & + w_{BadWords} \times BadWords(p.text) \\ & + w_{MisspelledWords} \times Misspelling(p.text) \end{aligned}$$

where

- $IsSPAM(p.text)$  is a SPAM detector that determines the probability  $\in [0, 100]$  of  $p.text$  being spam;
- $BadWords(p.text)$  measures the bad words proportion  $\in [0, 100]$  against the number of words in a text;
- $Misspelling(p.text)$  measures the misspelling errors proportion  $\in [0, 100]$ ;
- $w_{SPAM}$ ,  $w_{BadWords}$ , and  $w_{MisspelledWords}$  represent user-defined parameters to indicate the weights that the user gives to each filter, such that  $w_{SPAM} + w_{BadWords} + w_{MisspelledWords} = 1$ .

##### B. USER CREDIBILITY

User credibility analyzes only the user as a unit of the platform, without being influenced by other users, as it is described in Definition 2.

**Definition 2: User Credibility (UserCred).** Given a set of metadata of a user who published a post  $p$ , represented as  $p.user$ , User Credibility is a function, denoted as  $UserCred(p.user)$ , that returns a measure  $\in [0, 100]$ , defined as:

$$UserCred(p.user) = Verif(p.user) + Creation(p.user)$$

where:

- $Verif(p.user)$  is a function that returns 50 if the user is verified and 0 otherwise;
- $Creation(p.user)$  measures the time since the user's account was created, with a value between 0 and 50, increasing with the longevity of the account, such as  $Creation(p.user) = \frac{AccountAge(p.user)}{MaxAccountAge(p.platform)} \times 50$  where:
  - $AccountAge(p.user) = currentYear - YearJoined(p.user)$ ;
  - $MaxAccountAge(p.platform) = currentYear - SocialPlatformCreationYear(p.platform)$ ;

- $SocialPlatformCreationYear(p.platform)$  is the year in which the targeted social platform ( $p.platform$ ) was created (e.g., 2006 for Twitter).

### C. SOCIAL CREDIBILITY

Social credibility is focused on the relations between a user account and the other accounts on the social media platform. It considers the number of *followers* and *following* (friends), as shown in Definition 3.

**Definition 3: Social Credibility (SocialCred).** Given a set of metadata of a user who published a post  $p$ , represented as  $p.user$ , Social Credibility is a function, denoted as  $SocialCred(p.user)$ , that returns a measure  $\in [0, 100]$ , defined as:

$$SocialCred(p.user) = FollowersImpact(p.user) + FFProportion(p.user)$$

where:

- $FollowersImpact(p.user) = \frac{\min(p.user.followers, \max\_FOLLOWERS)}{\max\_FOLLOWERS} \times 50$  measures the impact  $\in [0, 50]$  on the number of followers;
- $FFProportion(p.user) = \frac{p.user.followers}{p.user.followers + p.user.following} \times 50$  measures the proportion  $\in [0, 50]$  between the number of followers and followings of the user.
- $\max\_FOLLOWERS$  is a user-defined parameter.

The  $\max\_FOLLOWERS$  constant is supplied by the user, for example, in [26] it is considered as 2 million.  $FFProportion$  is self-explanatory—a simple proportion that increases the credibility if the user has more *followers* than *followings*. The purpose of this function is to discredit bots, which tend to have more *followings* than *followers*. In this work, we reinforce this aspect at the User Credibility level, as we explain in Section V.

### D. GLOBAL CREDIBILITY LEVEL

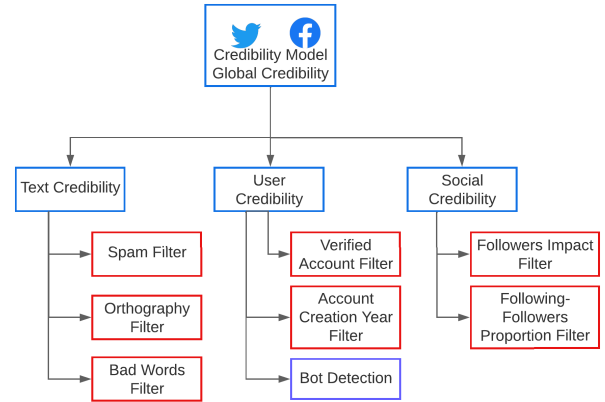
The credibility of a post is a weighted sum of the three credibility measures described previously. Definition 4 shows how it is calculated.

**Definition 4: Credibility Level (Cred).** Given a post,  $p$ , Credibility Level is a function, denoted as  $Cred(p)$ , that returns a measure  $\in [0, 100]$  of its level of credibility, defined as:

$$Cred(p) = w_{Text} \times TextCred(p.text) + w_{User} \times UserCred(p.user) + w_{Social} \times SocialCred(p.user),$$

where:

- $w_{Text}$ ,  $w_{User}$ , and  $w_{Social}$  represent the weights that the user gives to text credibility, user credibility, and social credibility, respectively, such that:  $w_{Text} + w_{User} + w_{Social} = 1$ ;
- $TextCred(p.text)$ ,  $UserCred(p.user)$ , and  $SocialCred(p.user)$  represent the credibility measure related to the text, the user, and the social impact of  $p$ , respectively.



**FIGURE 8.** T-Creo credibility model extended with the bot detection functionality (in purple).

Following section describes the use of bot detection into the credibility analysis.

### V. CREDIBOT: EXTENDED CREDIBILITY MODEL CONSIDERING BOT DETECTION

Identifying bots on Twitter gives to user more tools to analyze their perceptions about the shared data (credibility). As more processed information related to the tweets can be provided to users, the sharing can be better performed and being more careful. Bot detection classifies the Twitter accounts, thus the User Credibility category is the one to be modified in order to consider bot information in the credibility model. Figure 8 shows the different levels of the credibility model described in Section IV and the bot detection functionality added to User Credibility.

Users mainly associate bots to fake information disseminators, which according to the literature is not always true. Due to the existence of non-malicious bots (white bots), some assumptions are made for not hard penalization when the model classifies a user as a bot. The assumptions are presented as follows:

- If User Credibility value is higher than 50, then this credibility is penalized by 15%.
- If User Credibility value is between 35 and 50, then this credibility is penalized by 25%.
- If User Credibility value is less than 35, then this credibility is set to 0.

When the User Credibility value is higher than 50 (a low penalization is applied), it means it is either verified or ancient account; thus, the user could be a white bot. White bots are used to share true information mainly from government institutions. Definition 5 shows the redefinition of the User Credibility (Definition 2).

**Definition 5: User Credibility with bot detection (UserCredBot).** Given a set of metadata of a user who published a post  $p$ , represented as  $p.user$ , User Credibility with bot detection is a function, denoted as  $UserCredBot(p.user)$ ,

that returns a measure  $\in [0, 100]$ , defined as:

$$\text{UserCredBot}(p.\text{user}) = \begin{cases} \text{UserCred}(p.\text{user}) & \text{If } p.\text{user} \text{ is not a bot,} \\ 0.85 \times \text{UserCred}(p.\text{user}) & \text{If } p.\text{user} \text{ is a bot and} \\ & \text{UserCred}(p.\text{user}) \text{ is higher than 50,} \\ 0.75 \times \text{UserCred}(p.\text{user}) & \text{If } p.\text{user} \text{ is a bot and} \\ & \text{UserCred}(p.\text{user}) \text{ is between 35 and 50,} \\ 0 & \text{otherwise.} \end{cases}$$

where:

$\text{UserCred}(p.\text{user})$  is the original value given in Definition 2. Text and Social Credibility levels remain the same.

The new credibility level is formally defined in Definition 6.

**Definition 6: New Credibility Level ( $\text{Cred}'$ ).** Given a post,  $p$ , New Credibility Level is a function, denoted as  $\text{Cred}'(p)$ , that returns a measure  $\in [0, 100]$  of its level of credibility, defined as:

$$\begin{aligned} \text{Cred}'(p) = & w_{\text{Text}} \times \text{TextCred}(p.\text{text}) \\ & + w_{\text{User}} \times \text{UserCredBot}(p.\text{user}) \\ & + w_{\text{Social}} \times \text{SocialCred}(p.\text{user}) \end{aligned}$$

where:

- $w_{\text{Text}}$ ,  $w_{\text{User}}$ , and  $w_{\text{social}}$  represent the weights that the user gives to text credibility, user credibility, and social credibility, respectively, such as:  $w_{\text{Text}} + w_{\text{User}} + w_{\text{Social}} = 1$ ;
- $\text{TextCred}(p.\text{text})$ ,  $\text{UserCredBot}(p.\text{user})$ , and  $\text{SocialCred}(p.\text{user})$  represent the credibility measure related to the text, the user, and the social impact of  $p$ , respectively.

In the following section, the new credibility measure is experimentally validated.

## A. CREDIBILITY MODEL EVALUATION

To evaluate the extended credibility model, we implemented it with the Random Forest model, because of its better performance (see Section III) and integrated into T-CREo framework [27].

For the experiments, we designed two scenarios of tests. For the first scenario, fourteen Twitter accounts were selected, including 10 human and 4 bot accounts for each language. The human accounts correspond to accounts that have more followers, according to Wikipedia<sup>7</sup> and they are verified (7 verified accounts), and other accounts from genuine dataset (Genuine accounts - Table 3), which are not verified (3 not verified accounts). The bot accounts correspond to accounts known as bots from the experimentation dataset (spambot #1 - Table 3).

For the second scenario, 22 accounts were randomly selected from the fake followers dataset (Table 3), which were not used in the training or post-training validation process.

The results obtained from both scenarios are presented in the following sections. These results show the user credibility values for the original and extended model and their penalization in case that the user is qualified as a bot. Additionally, the “Verified Weight” and “Creation Weight” considered for both models, and the score obtained by Botometer for each account are also shown.

### 1) RESULTS FOR SCENARIO 1

Table 12 shows the results for accounts that publish tweets in English. There are two accounts, @ladygaga and @TheEllenShow, that were detected as ‘bot’ despite being recognized as genuine accounts by Botometer. Doing an analysis of these accounts on Twitter, for @ladygaga, her posts tend to promote events or products that correspond to her career, such as makeup or music, in addition of using at least one hashtag, mention, or URL in each post. This behavior could be taken as a spam bot.

Table 13 shows the accounts that publish tweets in Spanish. In this case, there are two “bot” accounts that do not lower their credibility, because their values for Verified Weight and Creation Weight were zero, respectively (as a consequence of the User Credibility definition – see Definition 2). Interestingly, in the case of the @RecuerdameBot account, a self-declared bot, was evaluated with a relatively high value by botometer and our model manages to classify it correctly. In this test, there are also two accounts that were detected as bot despite being labeled as genuine user (and verified as such also by Botometer). For @Rubiu5, this usually promotes events in which he participates using hashtags and some URLs, as well as promoting his own products, also commenting or retweeting publications of official accounts of organizations that have to do with video games, which could perhaps be taken as a behavior of bot. On the other hand, although the @metrodesantiago account corresponds to an official transportation account, it presents the behavior of a hybrid account, posting at certain times to indicate the start and end of train trips, or warning with the exact time about a problem on the track or in a station.

### 2) RESULTS FOR SCENARIO 2

For this test, the same number of accounts were selected for both Spanish and English and each account was analyzed to verify its validity (if it was active and maintained the same name), to ensure that the behavior remained the same that at the time of tagging. After verifying the validity of the 11 accounts for each language, the same analysis process was performed as in the first scenario regarding the calculation of the credibility of the original and the extended model. The results shown in Table 14 and Table 15 are consistent – i.e. considering the classification with Botometer, the credibility level according to our model, and the account label from the source.

In summary, the metrics of performance in the validation of bot detection included in the analysis of credibility

<sup>7</sup>[https://en.wikipedia.org/wiki/List\\_of\\_most-followed\\_Twitter\\_accounts](https://en.wikipedia.org/wiki/List_of_most-followed_Twitter_accounts)



**TABLE 12. Validation of analysis credibility including bot detection (English), Scenario 1.**

| user             | label | Botometer | credibility<br>(original)<br>(%) | credibility<br>(extended)<br>(%) | Detected as | Verified<br>Weight | Creation<br>Weight | Creation Date             |
|------------------|-------|-----------|----------------------------------|----------------------------------|-------------|--------------------|--------------------|---------------------------|
| @BarackObama     | human | 1.9       | 91.65                            | 91.65                            | human       | 50                 | 46.875             | 2007-03-05 22:08:25+00:00 |
| @NASA            | human | 2.3       | 92.01                            | 92.01                            | human       | 50                 | 46.875             | 2007-12-19 20:20:32+00:00 |
| @YouTube         | human | 1.6       | 87.49                            | 87.49                            | human       | 50                 | 34.375             | 2007-11-13 21:43:46+00:00 |
| @nytimes         | human | 3.4       | 92.00                            | 92.00                            | human       | 50                 | 46.875             | 2007-03-02 20:41:42+00:00 |
| @elonmusk        | human | 1.2       | 89.57                            | 89.57                            | human       | 50                 | 40.625             | 2009-06-02 20:12:29+00:00 |
| @ladygaga        | human | 0.6       | 80.26                            | 75.62                            | bot         | 50                 | 43.75              | 2008-03-26 22:37:48+00:00 |
| @TheEllenShow    | human | 1.2       | 90.81                            | 86.17                            | bot         | 50                 | 43.75              | 2008-08-14 03:50:42+00:00 |
| @IGisellePizarro | human | 0.6       | 36.19                            | 36.15                            | human       | 0                  | 34.375             | 2011-02-22 04:37:34+00:00 |
| @INicoleromany   | human | 0.2       | 48.38                            | 48.38                            | human       | 0                  | 28.125             | 2013-09-05 20:52:02+00:00 |
| @AlekhandraKhan  | human | 0.4       | 51.84                            | 51.84                            | human       | 0                  | 28.125             | 2013-02-13 12:53:41+00:00 |
| @DennaMcsparran  | bot   | 5         | 27.41                            | 19.16                            | bot         | 0                  | 25                 | 2014-03-04 18:11:08+00:00 |
| @YukikoTretter   | bot   | 5         | 36.87                            | 28.62                            | bot         | 0                  | 25                 | 2014-03-02 10:38:13+00:00 |
| @RocheIAmaro     | bot   | 5         | 38.67                            | 30.42                            | bot         | 0                  | 25                 | 2014-02-20 22:28:03+00:00 |
| @ElyseKendell    | bot   | 4.4       | 25.83                            | 17.58                            | bot         | 0                  | 25                 | 2014-02-26 08:57:36+00:00 |

**TABLE 13. Validation of analysis credibility including bot detection (Spanish), Scenario 1.**

| user             | label | Botometer | credibility<br>(original)<br>(%) | credibility<br>(extended)<br>(%) | Detected as | Verified<br>Weight | Creation<br>Weight | Creation Date             |
|------------------|-------|-----------|----------------------------------|----------------------------------|-------------|--------------------|--------------------|---------------------------|
| @shakira         | human | 0.9       | 89.96                            | 89.96                            | human       | 50                 | 40.625             | 2009-06-03 17:38:07+00:00 |
| @andresiniesta8  | human | 2.1       | 89.92                            | 89.92                            | human       | 50                 | 40.625             | 2009-11-18 09:33:28+00:00 |
| @Mineduc         | human | 1.6       | 76.22                            | 76.22                            | human       | 50                 | 37.5               | 2010-05-05 22:03:42+00:00 |
| @biobio          | human | 3.8       | 90.29                            | 90.29                            | human       | 50                 | 43.75              | 2008-05-03 15:15:06+00:00 |
| @RAEinforma      | human | 3.4       | 87.79                            | 87.79                            | human       | 50                 | 34.375             | 2011-08-07 18:41:46+00:00 |
| @Rubiu5          | human | 1.2       | 76.48                            | 72.31                            | bot         | 50                 | 34.375             | 2011-10-25 21:37:48+00:00 |
| @metrodesantiago | human | 3         | 78.36                            | 73.88                            | bot         | 50                 | 40.625             | 2009-09-11 19:15:54+00:00 |
| @AdriUmbreon     | human | 0.1       | 44.04                            | 44.04                            | human       | 0                  | 25                 | 2014-02-26 15:03:49+00:00 |
| @AndyChatlani    | human | 1         | 38.66                            | 38.66                            | human       | 0                  | 34.375             | 2011-12-12 11:08:18+00:00 |
| @Armunho         | human | 0.5       | 54.87                            | 54.87                            | human       | 0                  | 37.5               | 2010-12-28 01:32:48+00:00 |
| @eugeniojaque8   | bot   | 4.7       | 39.58                            | 39.58                            | bot         | 0                  | 0                  | 2022-06-28 02:53:52+00:00 |
| @RecuerdameBot   | bot   | 2.3       | 48.35                            | 46.28                            | bot         | 0                  | 6.25               | 2020-04-26 16:51:44+00:00 |
| @123trabajo      | bot   | 4.8       | 53.82                            | 42.48                            | bot         | 0                  | 34.375             | 2011-03-04 15:39:40+00:00 |
| @4geeksmxOfi     | bot   | 3.9       | 28.79                            | 28.79                            | bot         | 0                  | 0                  | 2022-09-02 18:12:55+00:00 |

**TABLE 14. Validation of analysis credibility including bot detection (English), Scenario 2.**

| user             | label | Botometer | credibility<br>(original)<br>(%) | credibility<br>(extended)<br>(%) | Detected as | Verified<br>Weight | Creation<br>Weight | Creation Date             |
|------------------|-------|-----------|----------------------------------|----------------------------------|-------------|--------------------|--------------------|---------------------------|
| @CrazyPrize66244 | bot   | 5         | 37.11                            | 29.89                            | bot         | 0                  | 21.875             | 2015-05-26 08:13:25+00:00 |
| @eddy95680       | bot   | 4.1       | 45.08                            | 41.98                            | bot         | 0                  | 37.5               | 2010-02-24 08:00:21+00:00 |
| @faamkai         | bot   | 4.6       | 42.18                            | 39.08                            | bot         | 0                  | 37.5               | 2010-08-30 04:47:49+00:00 |
| @eyadharb        | bot   | 4.9       | 44.84                            | 33.50                            | bot         | 0                  | 34.375             | 2011-01-20 06:22:28+00:00 |
| @ferdinandoghene | bot   | 4.6       | 39.17                            | 28.86                            | bot         | 0                  | 31.25              | 2012-11-21 16:57:30+00:00 |
| @GennieRossi5062 | bot   | 4.7       | 54.29                            | 42.94                            | bot         | 0                  | 34.375             | 2011-02-18 01:38:16+00:00 |
| @Phatpete        | bot   | 4.4       | 32.07                            | 28.72                            | bot         | 0                  | 40.625             | 2009-04-21 10:42:17+00:00 |
| @Malakai64       | bot   | 4.6       | 28.45                            | 25.35                            | bot         | 0                  | 37.5               | 2010-03-10 16:18:17+00:00 |
| @WhiteHouse      | human | 2.4       | 78.76                            | 78.76                            | human       | 50                 | 6.25               | 2020-11-03 20:55:58+00:00 |
| @CBS6Albany      | human | 1.8       | 73.75                            | 73.75                            | human       | 50                 | 40.625             | 2009-01-14 18:44:15+00:00 |
| @Don_Reuters     | human | 0.2       | 61.80                            | 61.80                            | human       | 50                 | 37.5               | 2010-06-19 01:13:28+00:00 |

present a precision=1.0, recall=0.8462, F1 score=0.9167, and accuracy=0.92, for both scenarios.

### 3) DISCUSSION: BOT DETECTION IMPACT ON THE CREDIBILITY LEVEL AND EXECUTION TIME

Comparing the results between the original and extended models, the credibility values for genuine accounts remain the same, while the accounts that were classified as “bot” have lowered their credibility, according to the new user credibility definition. However, accounts detected as bots

whose user credibility value is already zero (recently created and unverified account) keep the same credibility value in the extended model as in the original model. For example, accounts @eugeniojaque8 and @4geeksmxOfi have low credibility in the original model due to a low user credibility, and remain with the same credibility value in the extended model.

In this work, we assume that bots are potentially malicious and distribute false information. However, a more detailed evaluation about the type of bots is required for better

**TABLE 15. Validation of analysis credibility including bot detection (Spanish), Scenario 2.**

| user           | label | Botometer | credibility (original) (%) | credibility (extended) (%) | Detected as | Verified Weight | Creation Weight | Creation Date             |
|----------------|-------|-----------|----------------------------|----------------------------|-------------|-----------------|-----------------|---------------------------|
| @mariachua16   | bot   | 4.6       | 41.14                      | 38.05                      | bot         | 0               | 37.5            | 2010-03-12 23:56:19+00:00 |
| @gato2710      | bot   | 4.4       | 40.88                      | 37.78                      | bot         | 0               | 37.5            | 2010-02-15 15:49:58+00:00 |
| @ruthyohanna   | bot   | 2.8       | 55.28                      | 52.18                      | bot         | 0               | 37.5            | 2010-05-16 19:51:50+00:00 |
| @bevega        | bot   | 2.8       | 43.80                      | 32.45                      | bot         | 0               | 34.375          | 2011-04-06 15:35:32+00:00 |
| @famacali      | bot   | 3.3       | 45.21                      | 41.86                      | bot         | 0               | 40.625          | 2009-09-11 20:18:39+00:00 |
| @chussin_20    | bot   | 4.6       | 56.35                      | 53.00                      | bot         | 0               | 40.625          | 2009-11-05 18:41:49+00:00 |
| @mIrDy_23      | bot   | 2.1       | 55.75                      | 52.40                      | bot         | 0               | 40.625          | 2009-12-13 04:53:11+00:00 |
| @JulioCChagoya | bot   | 4.8       | 41.81                      | 38.45                      | bot         | 0               | 40.625          | 2009-12-27 02:57:23+00:00 |
| @yalid12       | human | 0         | 37.15                      | 37.15                      | human       | 50              | 25              | 2014-10-20 07:11:41+00:00 |
| @SpainUN       | human | 1.4       | 71.49                      | 71.49                      | human       | 50              | 37.5            | 2010-10-28 19:28:07+00:00 |
| @epineyro_ok   | human | 1.6       | 70.71                      | 70.71                      | human       | 50              | 34.375          | 2011-03-02 20:53:38+00:00 |

**TABLE 16. Execution time for the credibility analysis (Scenario 1).**

| user             | time (original credibility model) | time (extended credibility model) |
|------------------|-----------------------------------|-----------------------------------|
| @BarackObama     | 1041 ms                           | 1091 ms                           |
| @NASA            | 1223 ms                           | 1261 ms                           |
| @YouTube         | 910 ms                            | 950 ms                            |
| @nytimes         | 904 ms                            | 945 ms                            |
| @elonmusk        | 930 ms                            | 966 ms                            |
| @ladygaga        | 1048 ms                           | 1085 ms                           |
| @TheEllenShow    | 857 ms                            | 893 ms                            |
| @1GisellePizarro | 1044 ms                           | 1083 ms                           |
| @1Nicoleromany   | 842 ms                            | 885 ms                            |
| @AlekhandraKhan  | 840 ms                            | 885 ms                            |
| @DennaMcsparran  | 826 ms                            | 863 ms                            |
| @YukikoTretter   | 843 ms                            | 880 ms                            |
| @RochelAmaro     | 854 ms                            | 895 ms                            |
| @ElyseKendell    | 900 ms                            | 942 ms                            |
| @shakira         | 1125 ms                           | 1140 ms                           |
| @andresiniesta8  | 917 ms                            | 932 ms                            |
| @Mineduc         | 912 ms                            | 927 ms                            |
| @biobio          | 944 ms                            | 960 ms                            |
| @RAEinforma      | 909 ms                            | 924 ms                            |
| @Rubiu5          | 1101 ms                           | 1116 ms                           |
| @metrodesantiago | 840 ms                            | 855 ms                            |
| @AdriUmbreon     | 918 ms                            | 933 ms                            |
| @AndyChatlani    | 931 ms                            | 946ms                             |
| @Armunho         | 982 ms                            | 997 ms                            |
| @eugeniojaque8   | 946 ms                            | 960 ms                            |
| @RecuerdameBot   | 1316 ms                           | 1340 ms                           |
| @123trabajo      | 1017 ms                           | 1031 ms                           |
| @4geeksmxOfi     | 1044 ms                           | 1059 ms                           |

credibility analysis, since some institutions use bots to disseminate periodic information about events, such as earthquakes, traffic, weather, among others. In our experiments, the highest original credibility value for bot accounts is 56.35 and 53 with the extended credibility model, which might represent an indication that the detection of bots on Twitter could be performed through the proposed credibility metrics. For instance, we could consider as a bot, accounts which have a credibility value less than 50.

In addition and due to the requirement of maintain a real time processing, the run times obtained by both credibility models are also shown in Table 16 and Table 17. According to these results, the extended credibility model increments, in average, the execution time (with respect to the original

**TABLE 17. Execution time for the credibility analysis (Scenario 2).**

| user             | time (original credibility model) | time (extended credibility model) |
|------------------|-----------------------------------|-----------------------------------|
| @CrazyPrize66244 | 1043 ms                           | 1081 ms                           |
| @eddy95680       | 989 ms                            | 1028 ms                           |
| @faamkai         | 984 ms                            | 1022 ms                           |
| @eyadharb        | 970 ms                            | 1008 ms                           |
| @ferdinandoghene | 980 ms                            | 1020 ms                           |
| @GennieRossi5062 | 982 0ms                           | 1030 ms                           |
| @Phatpete        | 1002 ms                           | 1039 ms                           |
| @Malakai64       | 904 ms                            | 942 ms                            |
| @WhiteHouse      | 906 ms                            | 949 ms                            |
| @CBS6Albany      | 890 ms                            | 928 ms                            |
| @Don_Reuters     | 916 ms                            | 953 ms                            |
| @mariachua16     | 898 ms                            | 912 ms                            |
| @gato2710        | 888 ms                            | 907 ms                            |
| @ruthyohanna     | 906 ms                            | 922 ms                            |
| @bevega          | 899 ms                            | 956 ms                            |
| @famacali        | 916 ms                            | 956 ms                            |
| @chussin_20      | 919 ms                            | 936 ms                            |
| @mIrDy_23        | 877 ms                            | 897 ms                            |
| @JulioCChagoya   | 916 ms                            | 957 ms                            |
| @yalid12         | 921 ms                            | 936 ms                            |
| @SpainUN         | 899 ms                            | 937 ms                            |
| @epineyro_ok     | 904 ms                            | 919 ms                            |

one) by 27.86 ms for scenario 1 (Table 16) and 30 ms for scenario 2 (Table 17). These averages correspond to the total sum of the time difference of the credibility calculation of the extended and original model, divided by the total number of accounts analyzed (in this case, 28 for scenario 1 and 22 for scenario 2). These results confirm that the extended credibility model computation times are maintained in real time.

## VI. CONCLUSION AND FUTURE WORK

In this work, we show the process carried out for the development of a real-time bot detection functionality, with the intention of extending an existing credibility model for Twitter. Thus, the extended credibility model is able to consider the existence of bots in social networks as an important factor when performing the credibility analysis process. Some machine learning algorithms, such as Decision Tree, Random Forest, Bagging, Adaboost, Logistic

Regression, Support Vector Machine, K-Nearest Neighbor, and MultiLayer Perceptron are considered for evaluation and comparison. The selected classifier is Random Forest for both English and Spanish datasets with performance metrics of accuracies over 97%. This selection correspond to the capacity of the classifier for recognizing bots (true negative values), even more the possibility to provide a more interpretable result. Different tests show that the most important features for bot detection are favorites\_count, statuses\_count, followers\_count, friends\_count, and listed\_count, which are social features. Validation tests of the bot detection included in the new credibility analysis model are performed, yielding a performance of precision 1.0, recall 0.8462, F1-score 0.9167, and accuracy 0.92 for both English and Spanish model. To evaluate the performance of the extended model, it was incorporated into T-CREO, a framework for credibility analysis in Twitter in real time. In addition, the times are measured and it is found that the credibility analysis process is done in 1.26 seconds for the longest time.

Future work includes the expansion of the categories of recognized bots to include white bots and cyberbots. An interesting aspect to study is the behavior of bots within their network, such as expansion times, expansion patterns, activation times associated with topics, and to add these new features to the study. The exploration of hybrid methods with social network analysis in conjunction with machine learning algorithms remains to be investigated. Even though our main objective is to show the feasibility of extending our previous credibility model including bot detection, other machine learning methods such as deep learning rest to explore in the future. It could include by example, bigger datasets with different languages and additional features selected by different feature selection methods for bot detection. A next step of the project seeks improvements at the level of code optimization and making available the developed extension.

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